SRCNN on KV260: experiment evolution：Phase 2: architecture branches (packing, Winograd, scatter, tiling, DDR, combination)

Condensed from 231 registered experiments (results.csv). Only turning points are drawn; every ID links to experiments/<ID>_*/. Numbers are ms/frame; the evidence level is stated on each node.

Metric: ms/frame = cycles x clock period. HLS = HLS estimated cycles x target clock; RTL x routed = C/RTL co-simulation cycles x an independently closed post-route clock. No value is a board measurement.

Solid: baseline or parameter evolution, retry after fix, integration of an existing module; dashed: numerical evidence or inspiration, control, verification of the same design, replay. Edges are design lineage, not Git history.

passed, comparable metric (★ selected) · passed with warning · failed or below gate · invalid configuration · timeout or interrupted · planned, not run · analysis · partial experiment, no whole-design metric

E043 Fastest point of the sweep

W01 Winograd PE F(2x2,3x3)

Exact integer transforms; inverse divides by 4 exactly

PE only: exact

G03 Scatter with merged edge kernels

Replace gather by channel-reduced scatter of partial sums

HLS 44.1 ms

! Slower than the gather control G01 (29.8 ms); direction closed

S01 Intermediate maps in DDR

Expose the two scratch arrays through m_axi

✗ Global ports were not exported; tool still synthesized on-chip storage

S03 Output-tile fusion B=8..64

Fuse Conv1/Conv2 per output tile to shrink buffers

HLS 130 to 228 ms

✗ Recomputed halos cost more cycles than the buffers save

P01 Packing PE: two 8-bit products per DSP

27-bit packed multiply, gap 18, signed-low borrow repair

PE only: 1 DSP per pair, exact

W07 Winograd Conv1 in the network

Nine 3x3 pieces of the 9x9 kernel, conditional tile scan

HLS 22.9 ms

! 5.7 M Conv1 cycles: the conditional tile branch is taken at every position

P12 Packing in the whole network

Pair output channels so one DSP does two multiplications

HLS 5.42 ms

✗ DSP halved, but decoding and selectors need 292,053 LUT

W10 Continuous Winograd

Replace the conditional scan with a fixed-tile continuous pipeline

HLS 7.55 ms

! T16 only; the T8 variant W09 needs 135,496 LUT

W18 Winograd with narrow raw-weight bank

Keep only the transformed banks the MAC engines read

HLS 4.23 ms

! HLS timing warning; still slower than continuous direct convolution

C11 Winograd + packing combined

Stack the two DSP-saving routes

HLS 1.08 ms

✗ LUT 248,797 / FF 307,525: far over the device; gains do not add